

Multiple Choice (10 marks)

1. If the absolute temperature of a blackbody is increased by a factor of 3, the energy radiated per second per unit area does which of the following?
 - (a) Decreases by a factor of 81.
 - (b) Decreases by a factor of 9.
 - (c) Increases by a factor of 9.
 - (d) Increases by a factor of 81.
2. Which of the following is true about any system that undergoes a reversible thermodynamic process?
 - (a) There are no changes in the internal energy of the system.
 - (b) The temperature of the system remains constant during the process.
 - (c) The entropy of the system and its environment remains unchanged.
 - (d) The entropy of the system and its environment must increase.
3. An atom has filled $n = 1$ and $n = 2$ levels. How many electrons does the atom have?
 - (a) 2
 - (b) 4
 - (c) 6
 - (d) 10
4. If the total energy of a particle of mass m is equal to twice its rest energy, then the magnitude of the particle's relativistic momentum is
 - (a) $mc/2$
 - (b) $mc/(2^{1/2})$
 - (c) mc
 - (d) $(3^{1/2})mc$
5. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately
 - (a) $0.1 \text{ GeV}/c^2$
 - (b) $0.2 \text{ GeV}/c^2$
 - (c) $0.5 \text{ GeV}/c^2$
 - (d) $1.0 \text{ GeV}/c^2$

True / False (10 marks)

Answer True or False

6. True or False: In superconductors, the attraction between Cooper pairs arises from interactions with the ionic lattice.
7. True or False: Electromagnetic radiation emitted from a nucleus is most likely in the form of gamma rays.
8. True or False: For an electron in the $l = 2$ state, there are 5 allowed values of the magnetic quantum number m_l .
9. True or False: The negative muon, μ^- , *has properties that are most similar to those of the electron*.
9. True or False: Gas lasers operate by utilizing transitions between the energy levels of free atoms to produce laser light.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between internal energy and heat transfer in an ideal gas during a constant volume process, and explain why these quantities are equal in this scenario.
12. Discuss the allowed transitions for an electron in the $n = 4, l = 1$ state of hydrogen. Analyze which final states are accessible and explain why certain states cannot be reached.

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